that these equations reduce to the *Fourier transform* along the y-axis. This is done by setting σ to zero in the equations, and simplifying:

$$ReX(\sigma,\omega)\Big|_{\sigma=0} = \frac{2\sin(\omega)}{\omega} \qquad ImX(\sigma,\omega)\Big|_{\sigma=0} = 0$$

As illustrated in Fig. 32-3, these are the correct frequency domain signals, the same as found by directly taking the Fourier transform of the time domain waveform.

Strategy of the Laplace Transform

An analogy will help in explaining how the Laplace transform is used in signal processing. Imagine you are traveling by train at night between two cities. Your map indicates that the path is very straight, but the night is so dark you cannot see any of the surrounding countryside. With nothing better to do, you notice an altimeter on the wall of the passenger car and decide to keep track of the elevation changes along the route.

Being bored after a few hours, you strike up a conversation with the conductor: "Interesting terrain," you say. "It seems we are generally increasing in elevation, but there are a few interesting irregularities that I have observed." Ignoring the conductor's obvious disinterest, you continue: "Near the start of our journey, we passed through some sort of abrupt rise, followed by an equally abrupt descent. Later we encountered a shallow depression." Thinking you might be dangerous or demented, the conductor decides to respond: "Yes, I guess that is true. Our destination is located at the base of a large mountain range, accounting for the general increase in elevation. However, along the way we pass on the outskirts of a large mountain and through the center of a valley."

Now, think about how you understand the relationship between elevation and distance along the train route, compared to that of the conductor. Since you have directly measured the elevation along the way, you can rightly claim that you know *everything* about the relationship. In comparison, the conductor knows this same complete information, but in a simpler and more intuitive form: the location of the hills and valleys that *cause* the dips and humps along the path. While your description of the signal might consist of thousands of individual measurements, the conductor's description of the signal will contain only a few parameters.

To show how this is analogous to signal processing, imagine we are trying to understand the characteristics of some electric circuit. To aid in our investigation, we carefully measure the impulse response and/or the frequency response. As discussed in previous chapters, the impulse and frequency responses contain *complete* information about this linear system.